

Numerical Solution of the Radial Flow Diffusivity Equation for Carbon Dioxide Injection into a Reservoir

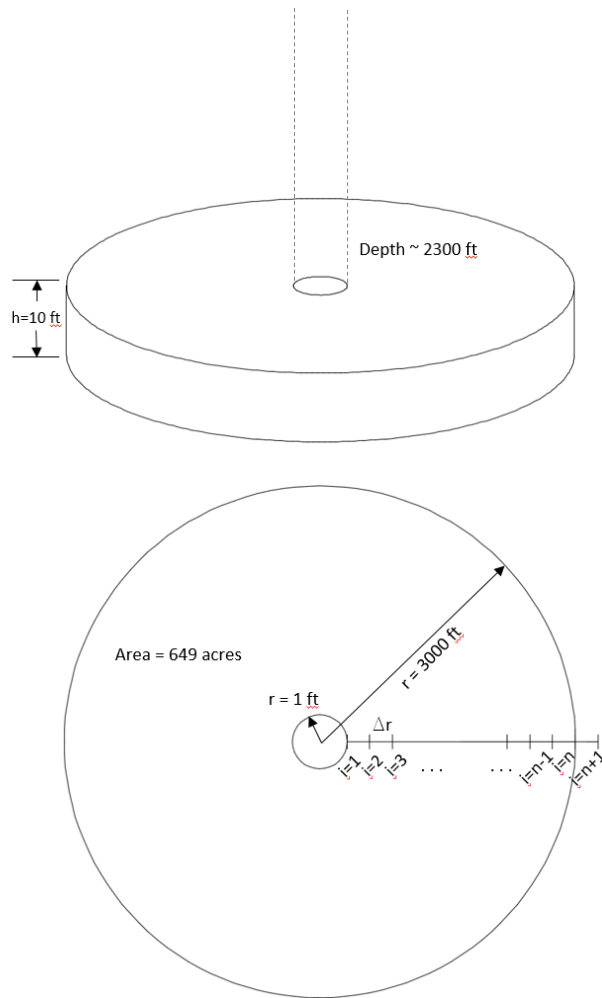
ENME 631 Numerical Methods for Engineers

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MODEL

Reservoir Shape, Dimensions, and Grid Nodes



Reservoir Parameters

Radius	3000 ft
Total area	649 acres
Wellbore radius	1 ft
Net Pay (height)	10 ft
Porosity	25%
Permeability	10 md
Gas Saturation	100%
Initial Pressure	100 psia
Temperature	100 °C
Formation Compressibility	$3 \times 10^{-6} \text{ psia}^{-1}$

Boundary Conditions

$t = 0$: $p = 100 \text{ psia}$ everywhere

Wellbore: $p = 1000 \text{ psia}$ (for all time)

Outer radius: no flow; $dp/dr = 0$ (for all time)

DIFFUSION EQUATION

Basic Differential Equation

$$\frac{0.234}{r} \frac{\partial}{\partial r} \left(0.001127 \frac{k}{\mu} \rho r \frac{\partial p}{\partial r} \right) = \rho \phi c_f \frac{\partial p}{\partial t} + \phi \frac{\partial p}{\partial t}$$

Darcy's Law

$$\longleftarrow v = -0.001127 \frac{k}{\mu} \frac{\partial p}{\partial r}$$

Introduce Real Gas Law and Compressibility

$$pV = znRT$$

$$C_g = \frac{1}{p} - \frac{1}{z} \frac{dz}{dp}$$

Diffusion Equation

$$\frac{\partial^2 m(p)}{\partial r^2} + \frac{1}{r} \frac{\partial m(p)}{\partial r} = \frac{\phi \mu c_t}{0.0002637 k} \frac{\partial m(p)}{\partial t}$$

Where,

$$m(p) = 2 \int_{p_r}^{p_1} \frac{p}{\mu z} dp$$

Diffusion Equation – Rearrange into Familiar Form

$$\frac{\partial m(p)}{\partial t} = a b \left(\frac{\partial^2 m(p)}{\partial r^2} + \frac{1}{r} \frac{\partial m(p)}{\partial r} \right)$$

Where,

$$a = \frac{0.0002637 k}{\phi}$$

$$b = \frac{1}{\mu c_t}$$

ALGEBRAIC EQUATIONS

Explicit Method

Finite Difference Equation

$$\frac{f_i^{n+1} - f_i^n}{\Delta t} = a b_i \left[\frac{f_{i-1}^n - 2f_i^n + f_{i+1}^n}{\Delta r^2} + \frac{1}{r_i} \left(\frac{f_{i+1}^n - f_{i-1}^n}{2\Delta r} \right) \right]$$

$$f_i^{n+1} = \omega b_i \left(1 - \frac{\Delta r}{2r_i} \right) f_{i-1}^n + (1 - 2\omega b_i) f_i^n + \omega b_i \left(1 - \frac{\Delta r}{2r_i} \right) f_{i+1}^n$$

Modified Equation

$$\frac{\partial f}{\partial t} = a b_i \left(\frac{\partial^2 f}{\partial r^2} + \frac{1}{r} \frac{\partial f}{\partial r} \right) - \frac{1}{2} \frac{\partial^2 f}{\partial t^2} \Delta t - \frac{1}{6} \frac{\partial^3 f}{\partial t^3} \Delta t^2 - \dots + \frac{a b}{6 r} \frac{\partial^3 f}{\partial r^3} \Delta r^2 + \dots$$

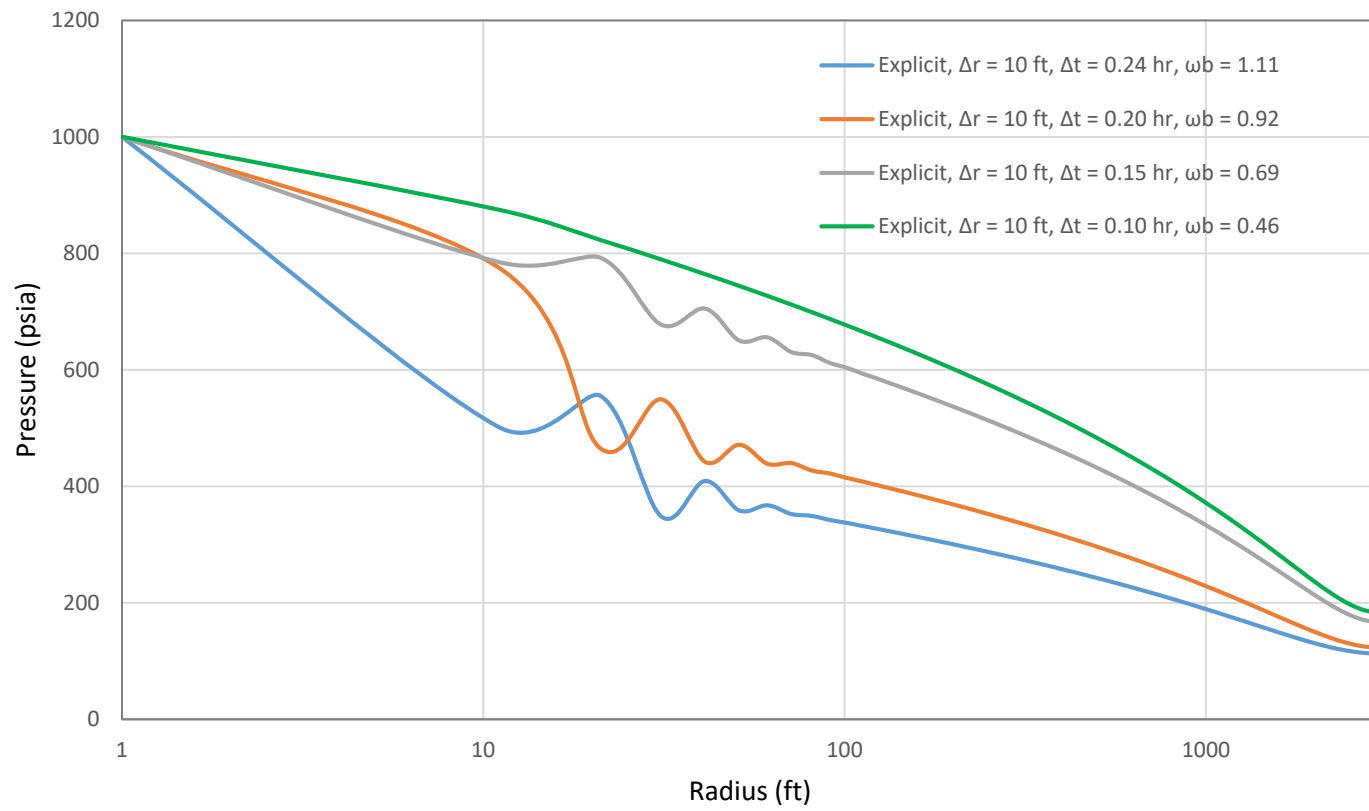
Stability Analysis

$$G = 1 + 2\omega b_i (\cos \theta - 1)$$

$$\omega b_i \leq \frac{1}{2}$$

NUMERICAL SOLUTION

Explicit Method Stability



Implicit Method

Finite Difference Equation

$$-\omega b_i \left(1 - \frac{\Delta r}{2r_i}\right) f_{i-1}^{n+1} + (1 + 2\omega b_i) f_i^{n+1} - \omega b_i \left(1 + \frac{\Delta r}{2r_i}\right) f_{i+1}^{n+1} = f_i^n$$

Matrix, Solved for Each Iteration in Time

$$\begin{bmatrix} 1 + 2\omega b_2 & -\omega b_2 \left(1 + \frac{\Delta r}{2r_2}\right) & 0 & \cdots & 0 & 0 & 0 \\ -\omega b_3 \left(1 - \frac{\Delta r}{2r_3}\right) & 1 + 2\omega b_3 & -\omega b_3 \left(1 + \frac{\Delta r}{2r_3}\right) & \cdots & 0 & 0 & 0 \\ 0 & -\omega b_4 \left(1 - \frac{\Delta r}{2r_4}\right) & 1 + 2\omega b_4 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -\omega b_{n-1} \left(1 - \frac{\Delta r}{2r_{n-1}}\right) & 1 + 2\omega b_{n-1} & -\omega b_{n-1} \left(1 + \frac{\Delta r}{2r_{n-1}}\right) \\ 0 & 0 & 0 & \cdots & 0 & -2\omega b_n & 1 + 2\omega b_n \end{bmatrix} \mathbf{x}$$

$$\begin{bmatrix} f_2^{n+1} \\ f_3^{n+1} \\ f_4^{n+1} \\ \vdots \\ f_{n-1}^{n+1} \\ f_n^{n+1} \end{bmatrix} = \begin{bmatrix} f_2^n + \omega b_2 \left(1 - \frac{\Delta r}{2r_2}\right) f_1^{n+1} \\ f_3^n \\ f_4^n \\ \vdots \\ f_{n-1}^n \\ f_n^n \end{bmatrix}$$

Crank-Nicolson Method

Finite Difference Equation

$$\begin{aligned}
 & -\omega b_i \left(1 - \frac{\Delta r}{2r_i}\right) f_{i-1}^{n+1} + 2(1 + 2\omega b_i) f_i^{n+1} - \omega b_i \left(1 + \frac{\Delta r}{2r_i}\right) f_{i+1}^{n+1} \\
 & = \omega b_i \left(1 - \frac{\Delta r}{2r_i}\right) f_{i-1}^n + 2(1 - \omega b_i) f_i^n + \omega b_i \left(1 + \frac{\Delta r}{2r_i}\right) f_{i+1}^n
 \end{aligned}$$

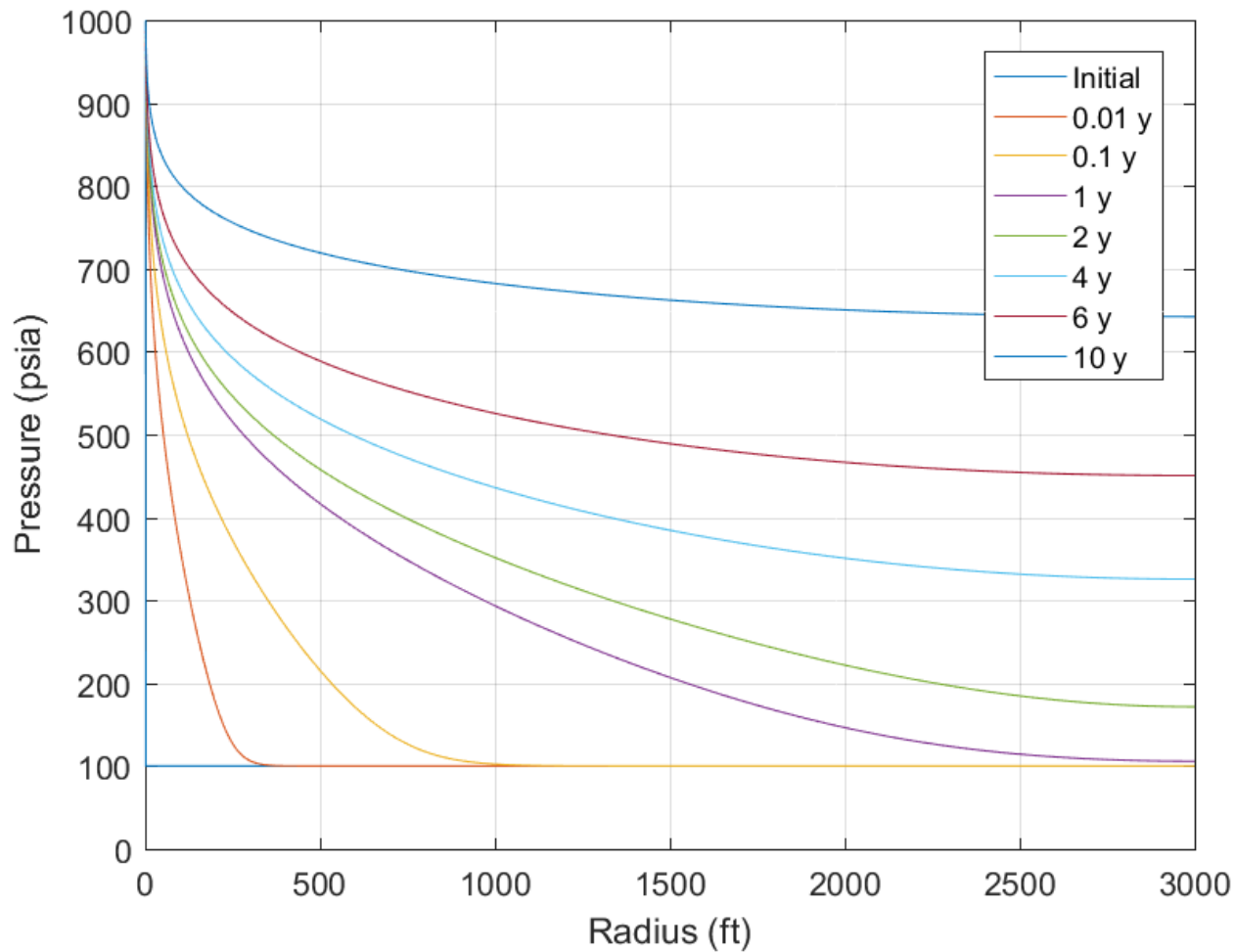
Matrix, Solved for Each Iteration in Time

$$\begin{bmatrix}
 2(1 + 2\omega b_2) & -\omega b_2(1 + \frac{\Delta r}{2r_2}) & 0 & \dots & 0 & 0 & 0 \\
 -\omega b_3(1 - \frac{\Delta r}{2r_3}) & 2(1 + 2\omega b_3) & -\omega b_3(1 + \frac{\Delta r}{2r_3}) & \dots & 0 & 0 & 0 \\
 0 & -\omega b_4(1 - \frac{\Delta r}{2r_4}) & 2(1 + 2\omega b_4) & \dots & 0 & 0 & 0 \\
 \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\
 0 & 0 & 0 & \dots & -\omega b_{n-1}(1 - \frac{\Delta r}{2r_{n-1}}) & 2(1 + 2\omega b_{n-1}) & -\omega b_{n-1}(1 + \frac{\Delta r}{2r_{n-1}}) \\
 0 & 0 & 0 & \dots & 0 & -2\omega b_n & 2(1 + 2\omega b_n)
 \end{bmatrix} \times$$

$$\begin{bmatrix} f_2^{n+1} \\ f_3^{n+1} \\ f_4^{n+1} \\ \vdots \\ f_{n-1}^{n+1} \\ f_n^{n+1} \end{bmatrix} = \begin{bmatrix} \omega b_2 \left(1 - \frac{\Delta r}{2r_2}\right) f_1^n + 2(1 - \omega b_2) f_2^n + \omega b_2 \left(1 + \frac{\Delta r}{2r_2}\right) f_3^n + \omega b_2 \left(1 - \frac{\Delta r}{2r_2}\right) f_1^{n+1} \\ \omega b_3 \left(1 - \frac{\Delta r}{2r_3}\right) f_2^n + 2(1 - \omega b_3) f_3^n + \omega b_3 \left(1 + \frac{\Delta r}{2r_3}\right) f_4^n \\ \vdots \\ \omega b_n \left(1 - \frac{\Delta r}{2r_n}\right) f_{n-1}^n + 2(1 - \omega b_n) f_n^n + \omega b_n \left(1 + \frac{\Delta r}{2r_n}\right) f_{n+1}^n \end{bmatrix}$$

NUMERICAL SOLUTION

Pressure profile—Implicit Method; Δt 1 hr; Δr 1 ft



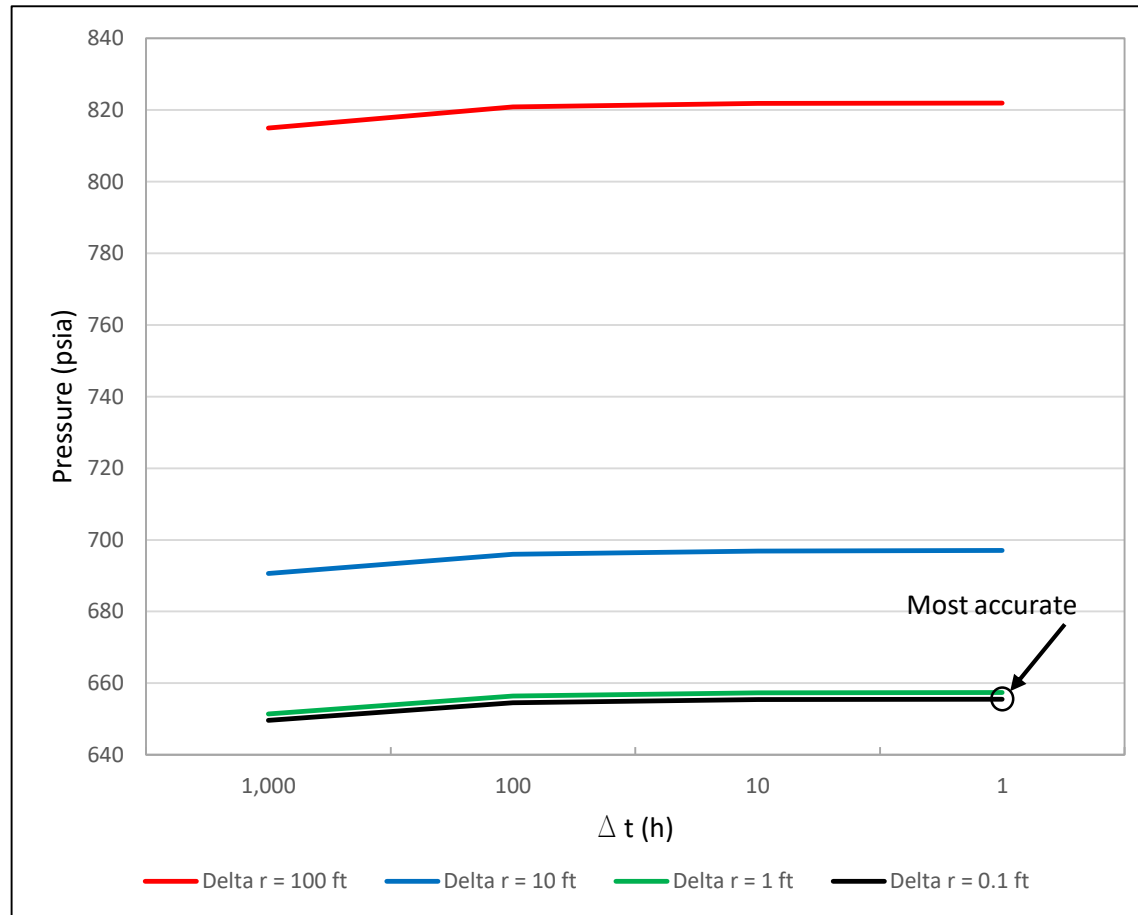
RESULTS

Solution for the Three Methods: Injection at 1000 psia Flowing Bottom Hole Pressure for 10 Years

Input Parameters				Average Reservoir Pressure (psia)			Run Time (s)		
Δt (h)	Δr (ft)	# Nodes	$\omega b_{\max}$	Explicit	Implicit	Crank-Nicolson	Explicit	Implicit	Crank-Nicolson
1,000	100	30	46.36	na	815.0	815.9	na	0.0	0.0
1,000	10	300	4,635.64	na	690.6	690.6	na	0.0	0.0
1,000	1	3,000	463,563.50	na	651.4	651.3	na	0.1	0.1
1,000	0.1	30,000	46,356,350.40	na	649.6	649.5	na	1.0	1.1
100	100	30	4.64	na	820.9	821.0	na	0.0	0.0
100	10	300	463.56	na	696.0	696.0	na	0.1	0.1
100	1	3,000	46,356.35	na	656.4	656.4	na	1.0	1.1
100	0.1	30,000	4,635,635.04	na	654.6	654.5	na	9.6	10.8
10	100	30	0.46	821.8	821.8	821.8	0.1	0.1	0.1
10	10	300	46.36	na	696.9	696.9	na	1.0	1.1
10	1	3,000	4,635.64	na	657.3	657.3	na	9.8	11.0
10	0.1	30,000	463,563.50	na	655.4	655.4	na	95.8	108.1
1	100	30	0.05	821.9	821.9	821.9	0.8	1.2	1.3
1	10	300	4.64	na	697.0	697.0	na	10.0	11.0
1	1	3,000	463.56	na	657.4	657.4	na	96.4	108.4
1	0.1	30,000	46,356.35	na	655.5	655.5	na	966.9	1,088.9
0.001	1	3000	0.46	657.4			69,717.7		

RESULTS

Implicit Method accuracy improvement with reducing values of Δt and Δr

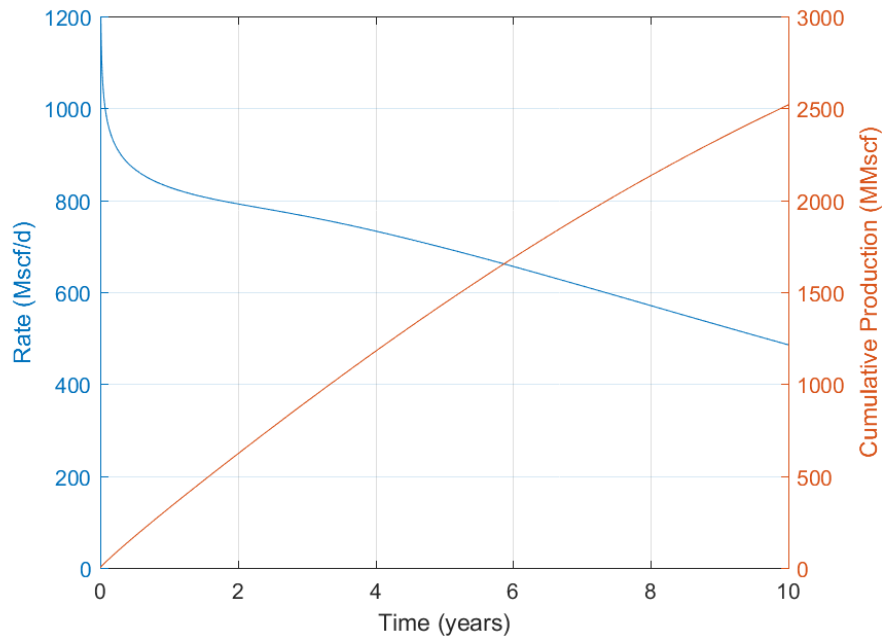


RESULTS

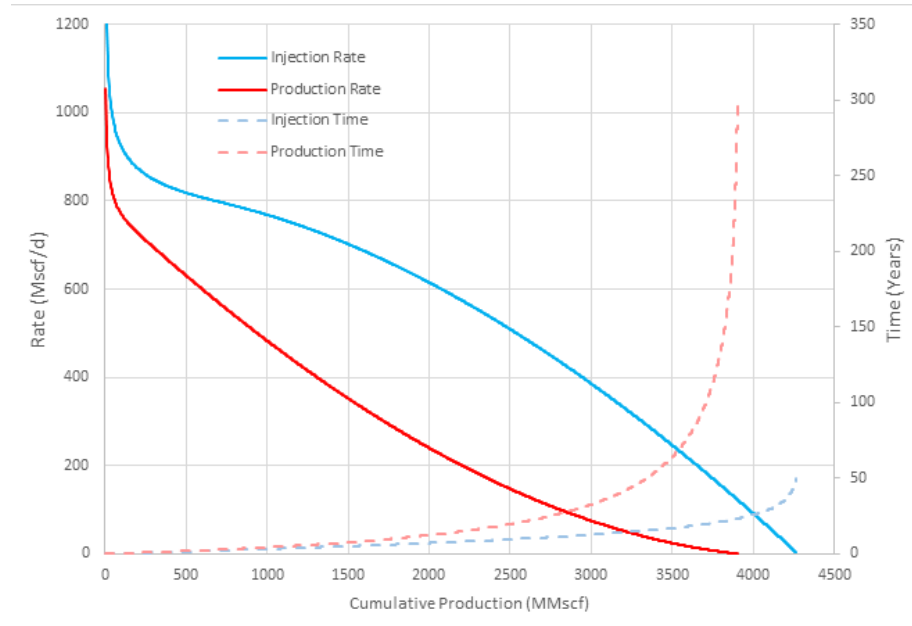
Deliverability

$$q = 0.01988 \frac{T_s k h (p_1^2 - p_2^2)}{p_s T (z\mu) \ln\left(\frac{r_2}{r_1}\right)}$$

Deliverability—CO₂ injection rate and cumulative injection vs. time; Implicit Method; Δt 1 hr; Δr 1 ft



Deliverability—CO₂ injection and blowdown; Implicit Method; Δt 100 hr; Δr 0.1 ft



CONCLUSIONS

Objectives set out in this project were achieved

- Demonstrated the application of numerical methods towards the solution of a practical engineering problem
- Insight obtained into to how CO₂ injection rate could be expected to vary over a 10 year time period

Further Efforts

- Implement a scaled Δr —Smaller near the wellbore, getting larger with increasing radius
- Crank-Nicolson Method solution without lagging non-linear term—solve the non-linear system of equations

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